Basic Principles of Medicine II

Module: Nervous System and Behavioral Sciences | Lecture NB 24

Pharyngeal Apparatus Derivatives; Neural Tube Defects

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Recommended Reading

- Gary C. Schoenwolf, PhD; Steven B. Bleyl, MD, PhD; Philip R. Brauer, PhD; Philippa H. Francis-West, PhD

Larsen's Human Embryology, 4, 78-103

- T.V.N. (Vid) Persaud, MD, PhD, DSc, FRCPath (Lond.), FAAA; Mark G. Torchia, MSc, PhD

The Developing Human, 17, 359-397

- Gary C. Schoenwolf, PhD; Steven B. Bleyl, MD, PhD; Philip R. Brauer, PhD; Philippa H. Francis-West, PhD

Larsen's Human Embryology, 17, 419-462

Objectives

SOM.MK.I.BPM2.1.NB.1.ANAT.0207	Describe the structures formed by each pharyngeal arch/groove/pouch that contributes to the formation of the face
SOM.MK.I.BPM2.1.NB.1.ANAT.0208	Describe the signs and symptoms of first arch syndrome
SOM.MK.I.BPM2.1.NB.1.ANAT.0209	Describe the development of the branchial sinuses, fistulas and cysts
SOM.MK.I.BPM2.1.NB.1.ANAT.0210	Describe the defects that can develop due to abnormal development of the grooves and pouches
SOM.MK.I.BPM2.1.NB.1.ANAT.0215	Describe the formation of the tongue, give the different arches that contribute and relate this to its innervation
SOM.MK.I.BPM2.1.NB.1.ANAT.0216	Describe the lymph drainage of the tongue and how it links to embryological development
SOM.MK.I.BPM2.1.NB.1.ANAT.0217	Describe the development of normal and ectopic thyroid and parathyroid glands
SOM.MK.I.BPM2.1.NB.1.ANAT.0218	Describe the development of a thyroglossal duct cyst and its clinical presentation
SOM.MK.I.BPM2.1.NB.1.ANAT.0219	Describe the development of the pituitary gland
SOM.MK.I.BPM2.1.NB.1.ANAT.0220	Describe the development of craniopharyngioma
SOM.MK.I.BPM2.1.NB.1.ANAT.0221	Describe the development of meroencephaly, anencephaly microcephaly and holoprosencephaly
SOM.MK.I.BPM2.1.NB.1.ANAT.0222	Describe the development of the Arnold-Chiari malformation
SOM.MK.I.BPM2.1.NB.1.ANAT.0223	Compare and contrast the cause of communicating versus noncommunicating hydrocephalus

Slides with a green background are recap or additional information and will not be discussed in lecture

Arch - Derivatives

Arch	Cranial nerve	Artery	Skeleton	Muscles
1st	Trigeminal (V2, V3)	Maxillary	Malleus, incus, spheno-mandibular ligament, maxilla, mandible, squamous temporal bone, zygomatic bone, portion of vomer	Mastication, tensor veli palatini, tensor tympani, anterior digastric, mylohyoid
2nd	Facial (V)		Stapes, styloid process, stylohyoid ligament, hyoid (lesser horn and upper body)	Stapedius, stylohyoid, posterior digastric
3rd	Glosso-pharyngeal (IX)	Common and internal carotid	Hyoid (greater horn and lower body)	Stylopharyngeus
4th	Vagus (X) - superior laryngeal	Aortic arch, right subclavian	Laryngeal cartilages	Pharyngeal constrictors, levator veli palatini, cricothyroid
6th	Vagus (X) - recurrent laryngeal	Ductus arteriosus, pulmonary arteries	Laryngeal cartilages	Intrinsic muscles of larynx, striated muscles of esophagus

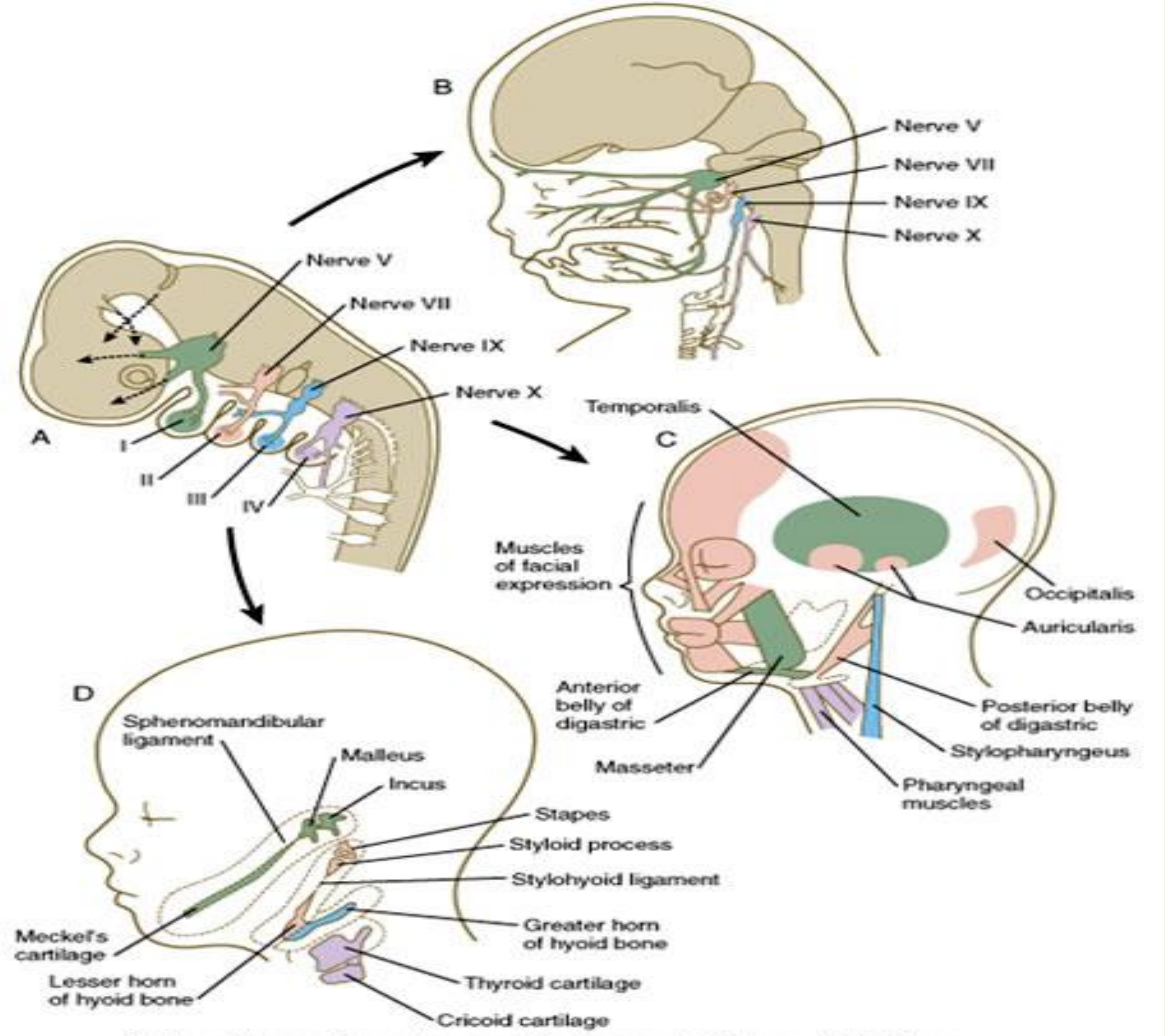
Derivatives of the grooves (clefts), pouches and membrane

GROOVES	
1st	External acoustic meatus
2nd	Obliterated by proliferation of 2nd arch to join the 6th arch
3rd	Obliterated by proliferation of 2nd arch to join the 6th arch
4th	Obliterated by proliferation of 2nd arch to join the 6th arch

POUCHES	
1st	Tubotympanic recess, middle ear cavity, mastoid antrum
2nd	Palatine tonsils, tonsillar fossa
3rd	Thymus, inferior parathyroid glands
4th	Superior parathyroid glands, ultimopharyngeal body (parafollicular cells of thyroid)

PHARYNGEAL MEMBRANE	
1st	Tympanic membrane

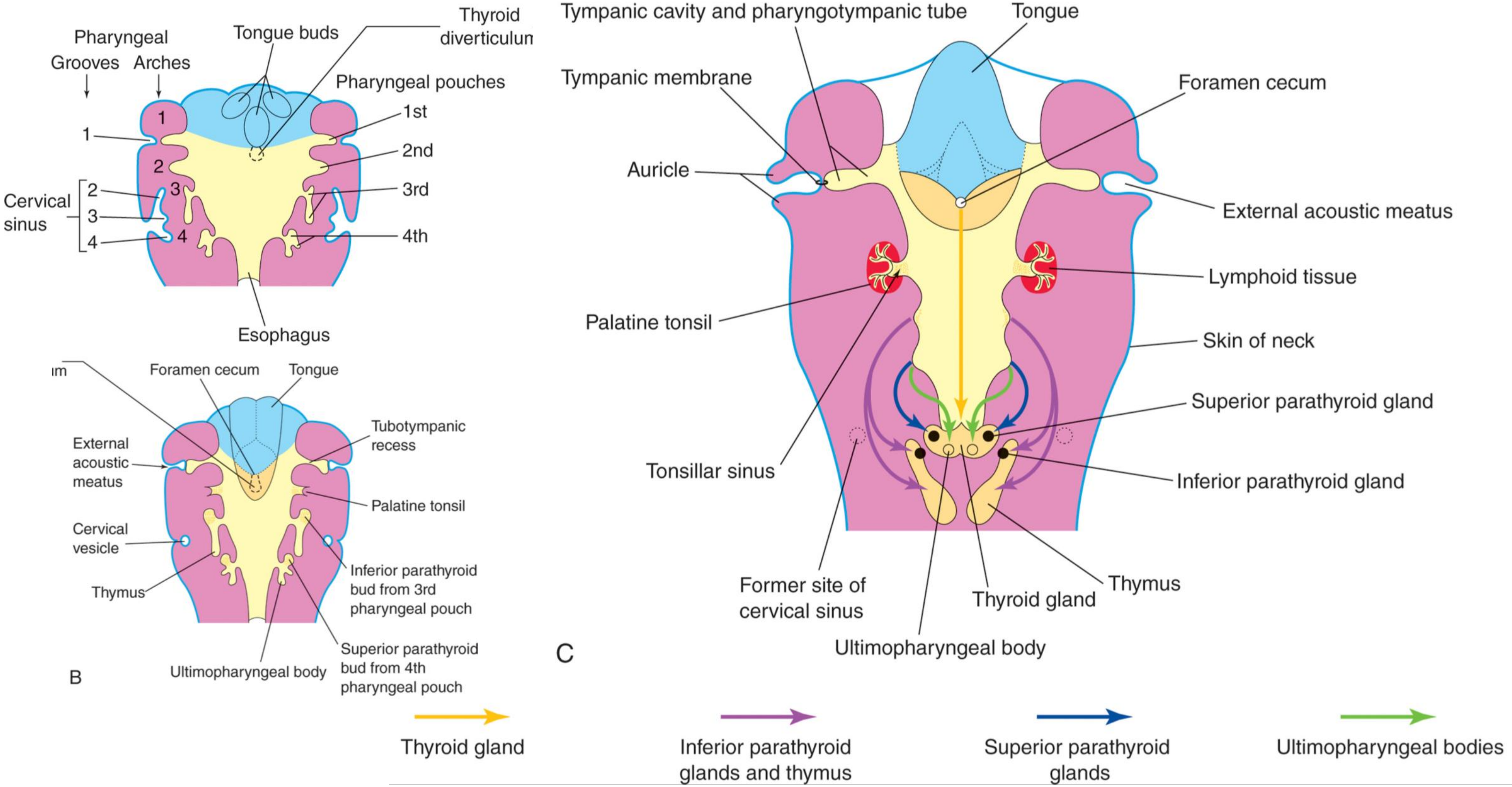
Pharyngeal arch derivatives



Carlson: Human Embryology and Developmental Biology, 4th Edition.

- 1[st] arch
- 2[nd] arch
- 3[rd] arch
- 4[th] arch

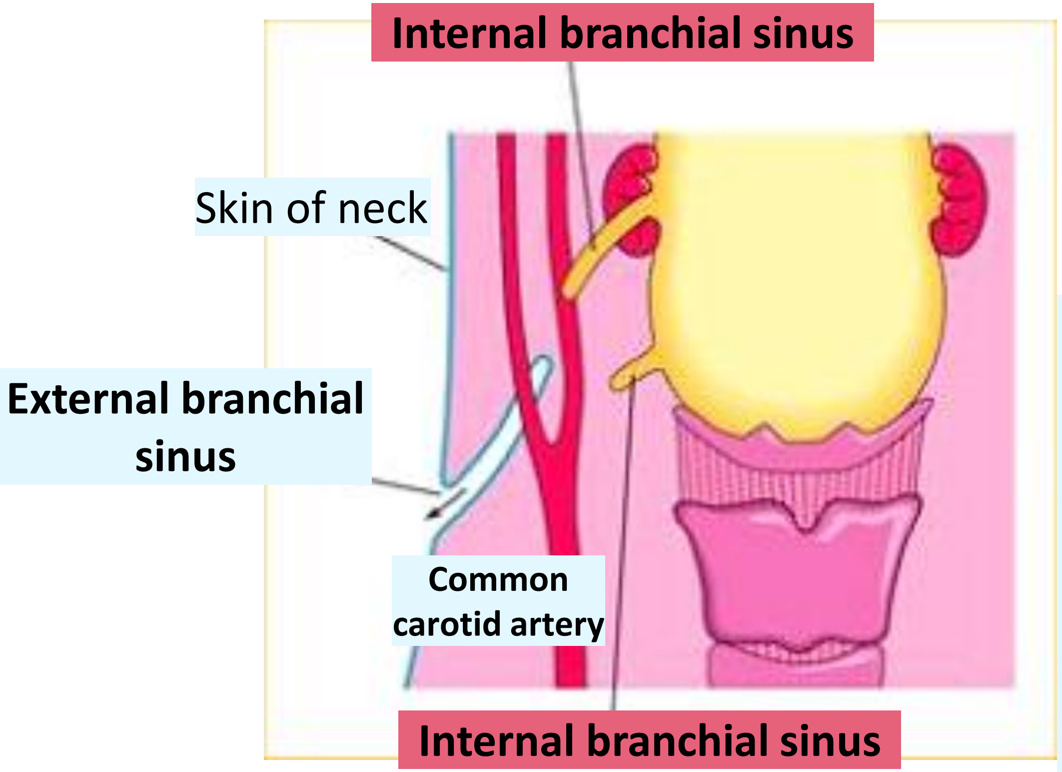
Neck development - Pharyngeal pouches and grooves

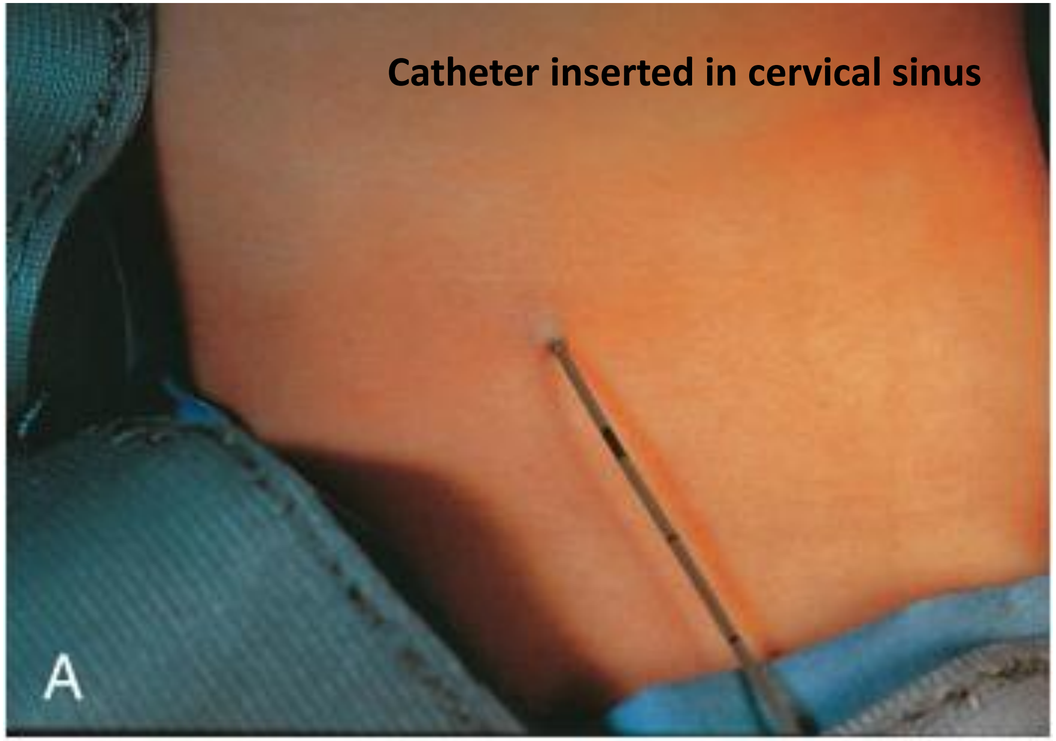


CERVICAL (BRANCHIAL) SINUS

External

- **Open externally along anterior border of SCM**
- Failure of the **second pharyngeal groove** and cervical sinus to obliterate



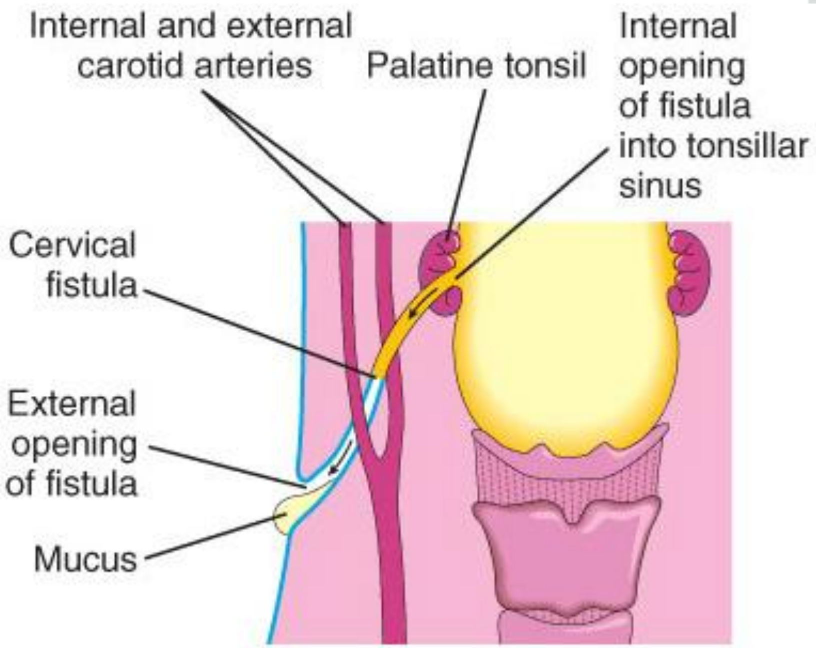


Internal

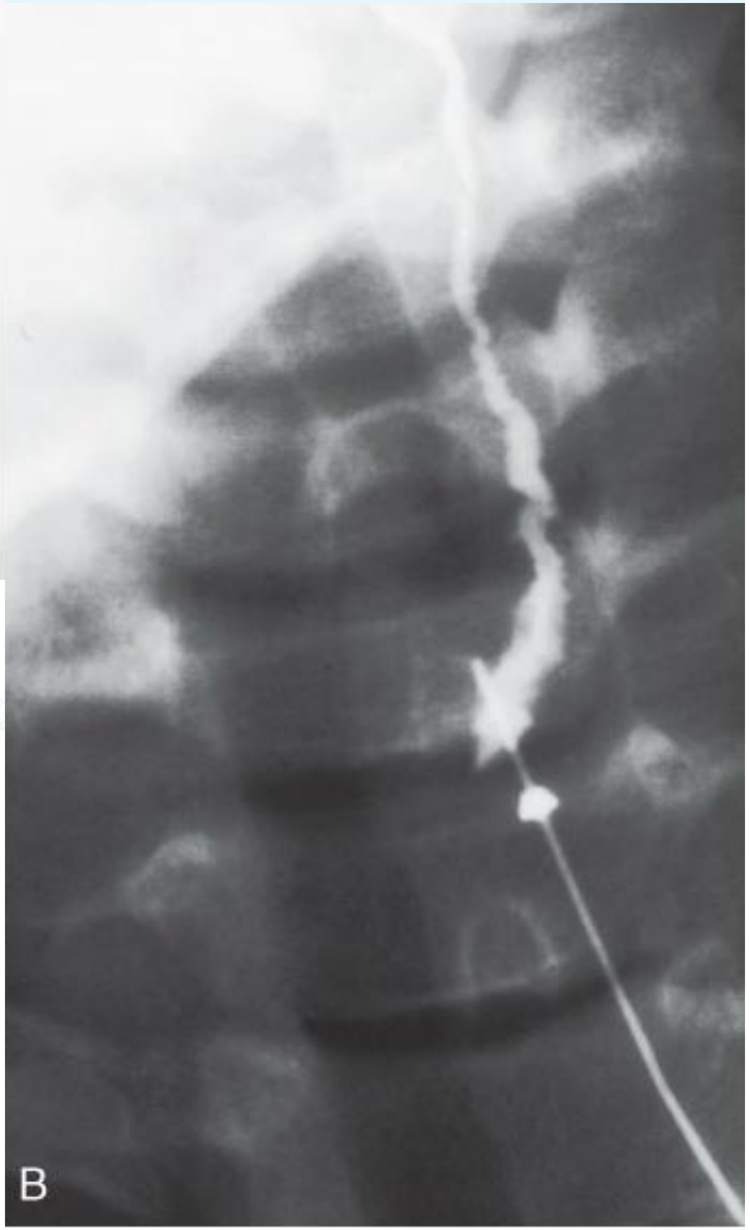
- **Opens internally into the tonsillar fossa** or near the palatopharyngeal arch
- These sinuses are rare. Most result from persistence of the proximal part of the **second pouch**.

CERVICAL (BRANCHIAL) FISTULA

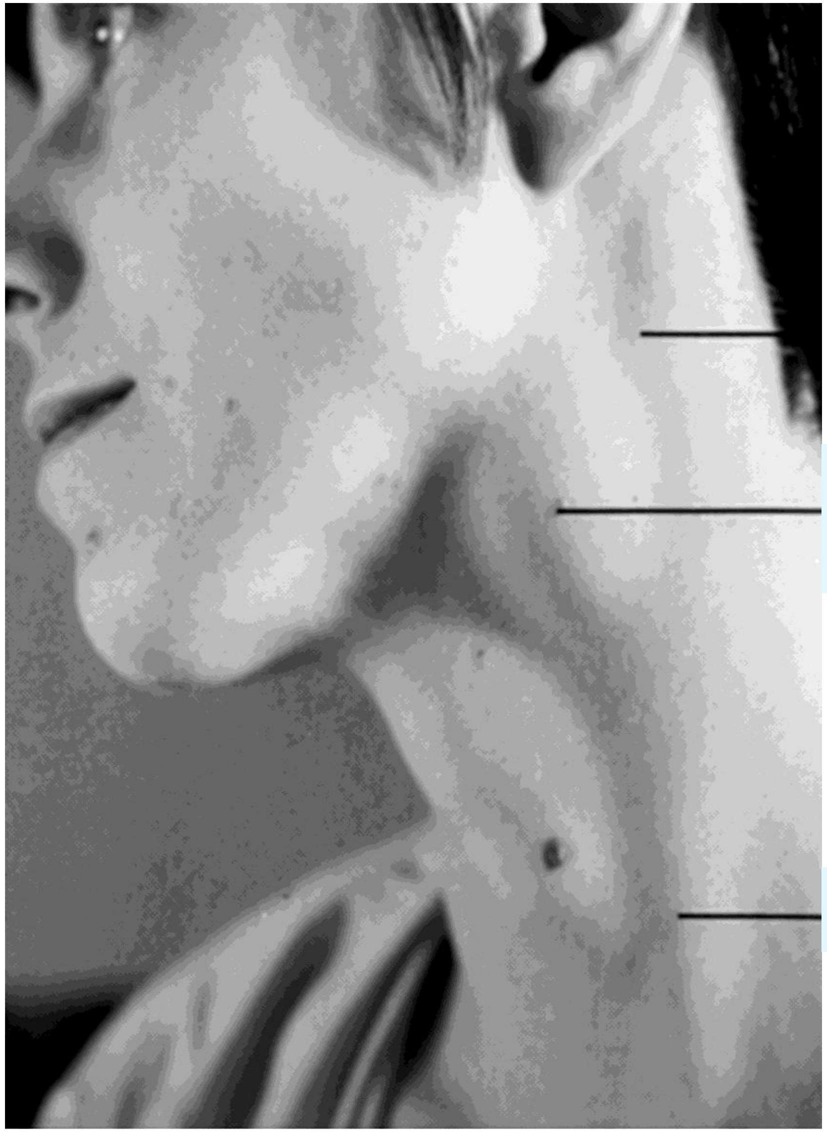
- Persistence of the **2nd pouch** and **cervical sinus**.
- Abnormal tract passing between external and internal carotid arteries
- Opening on the side of neck externally & in the tonsillar sinus internally.



Radiograph with contrast of a complete fistula

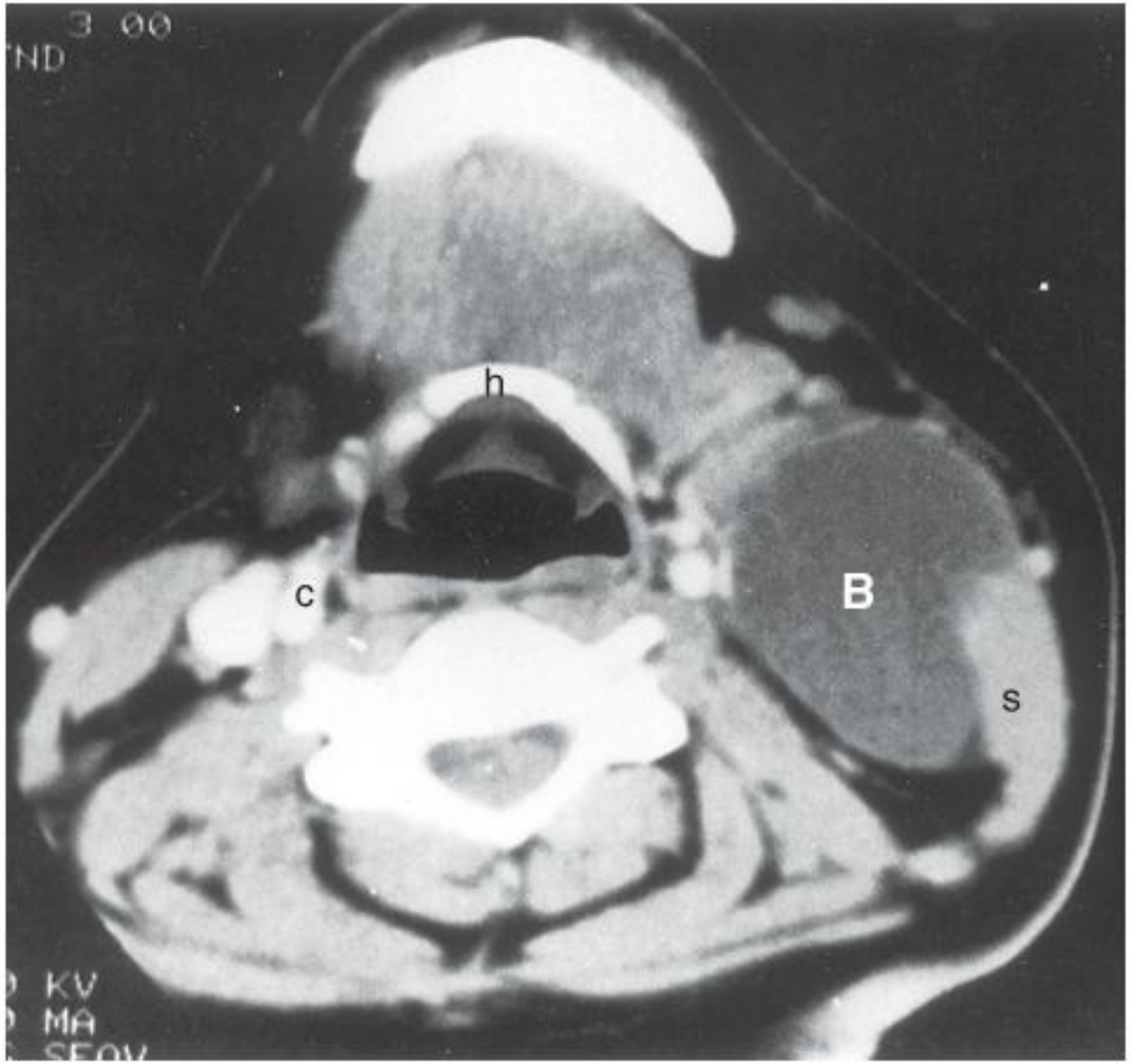


CERVICAL (BRANCHIAL) CYSTS

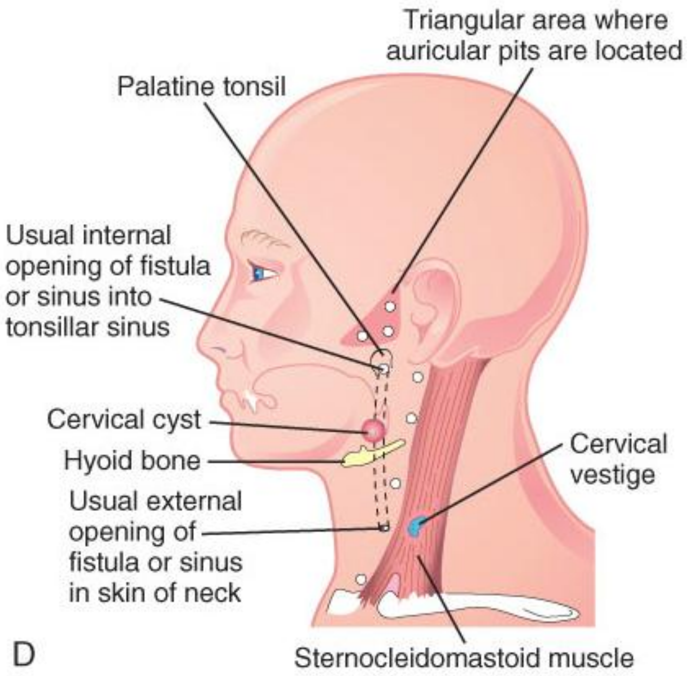


- Persistence of parts of **cervical sinus** and /or **2nd pharyngeal groove** with no internal or external communication
- **Cystic swelling along anterior border of the sternocleidomastoid**
- Slowly enlarges with fluid and desquamated cells
- Painless
- Seen in late childhood

CERVICAL (BRANCHIAL) CYST

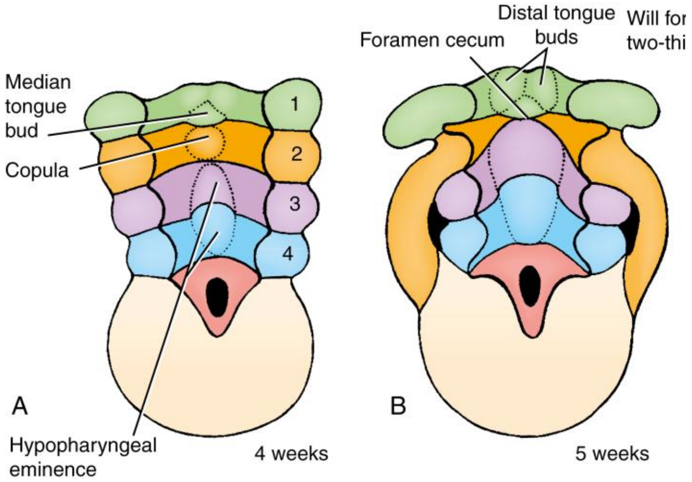


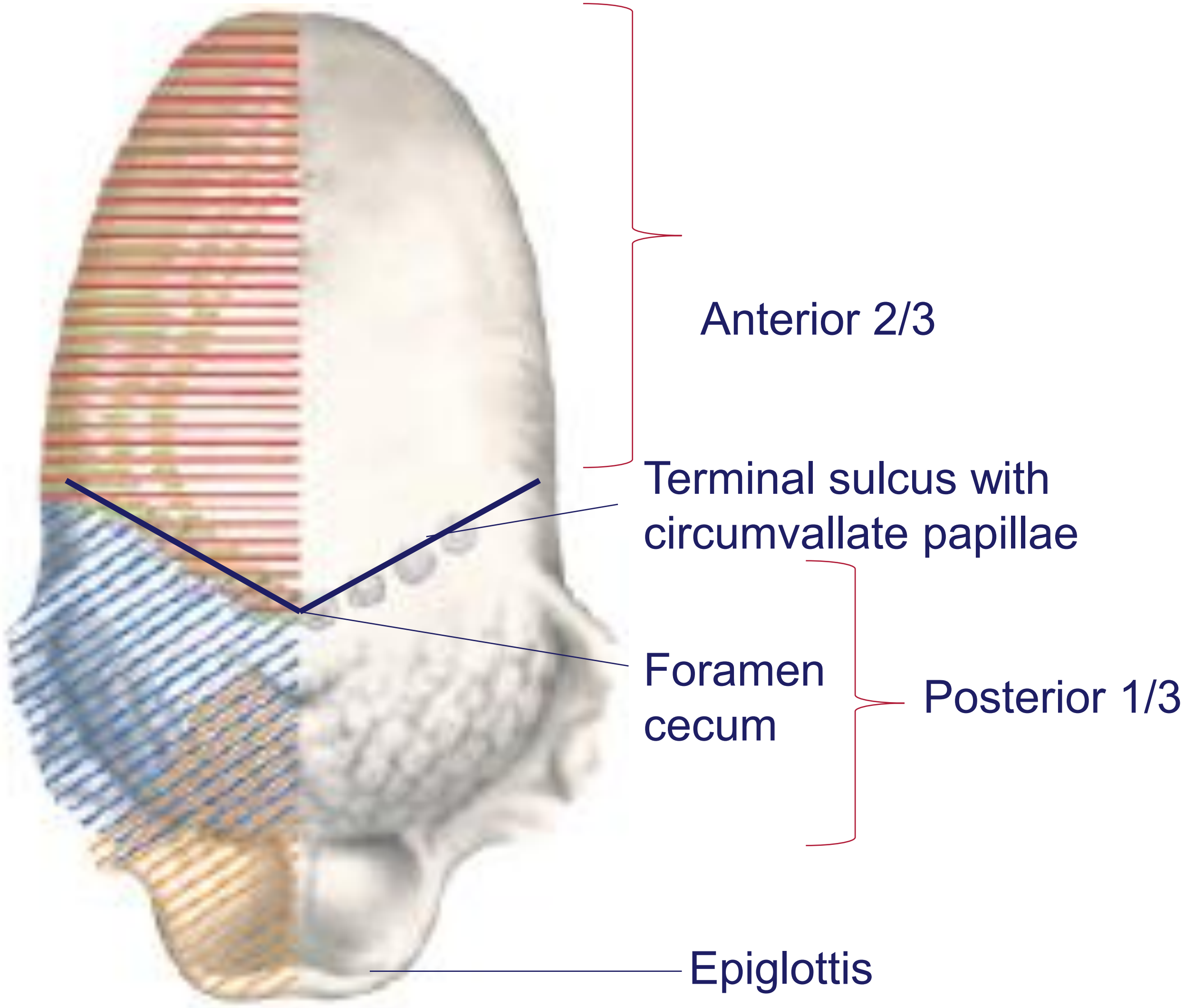
CT of large cervical cyst (**B**) anterior to SCM (**s**)



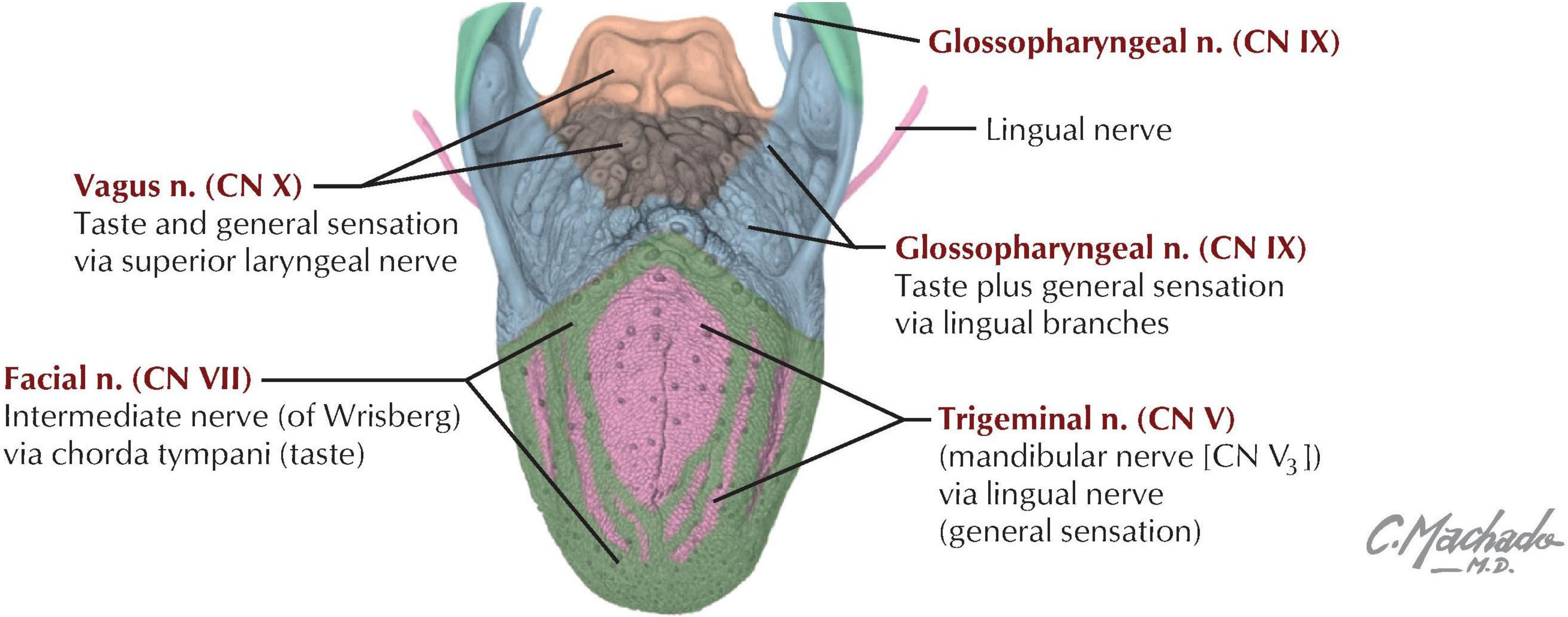
Development of the tongue

- Development of the tongue is from pharyngeal arches 1 to 4
- 1st arch
  - 2 lateral swellings (lingual swellings)
  - 1 median swelling (tuberculum impar)
- 2nd arch
  - Midline Copula
- 3rd and 4th arch
  - Hypobranchial eminence
- **Muscles** of the tongue form from myoblasts that migrate from **occipital somites**





Nerve supply of the tongue



Motor: Hypoglossal Nerve (CN XII), except for palatoglossus which is vagus (CN X)

First arch syndromes

Abnormal development of the first pharyngeal arch due to insufficient migration of neural crest

Treacher Collins syndrome (1:50 000)

- Autosomal dominant
- Micrognathia
- Cleft palate
- Down slanting eyes
- Low set ears (may have middle and inner ear deformities)
- Deafness
- Lower eyelid defects

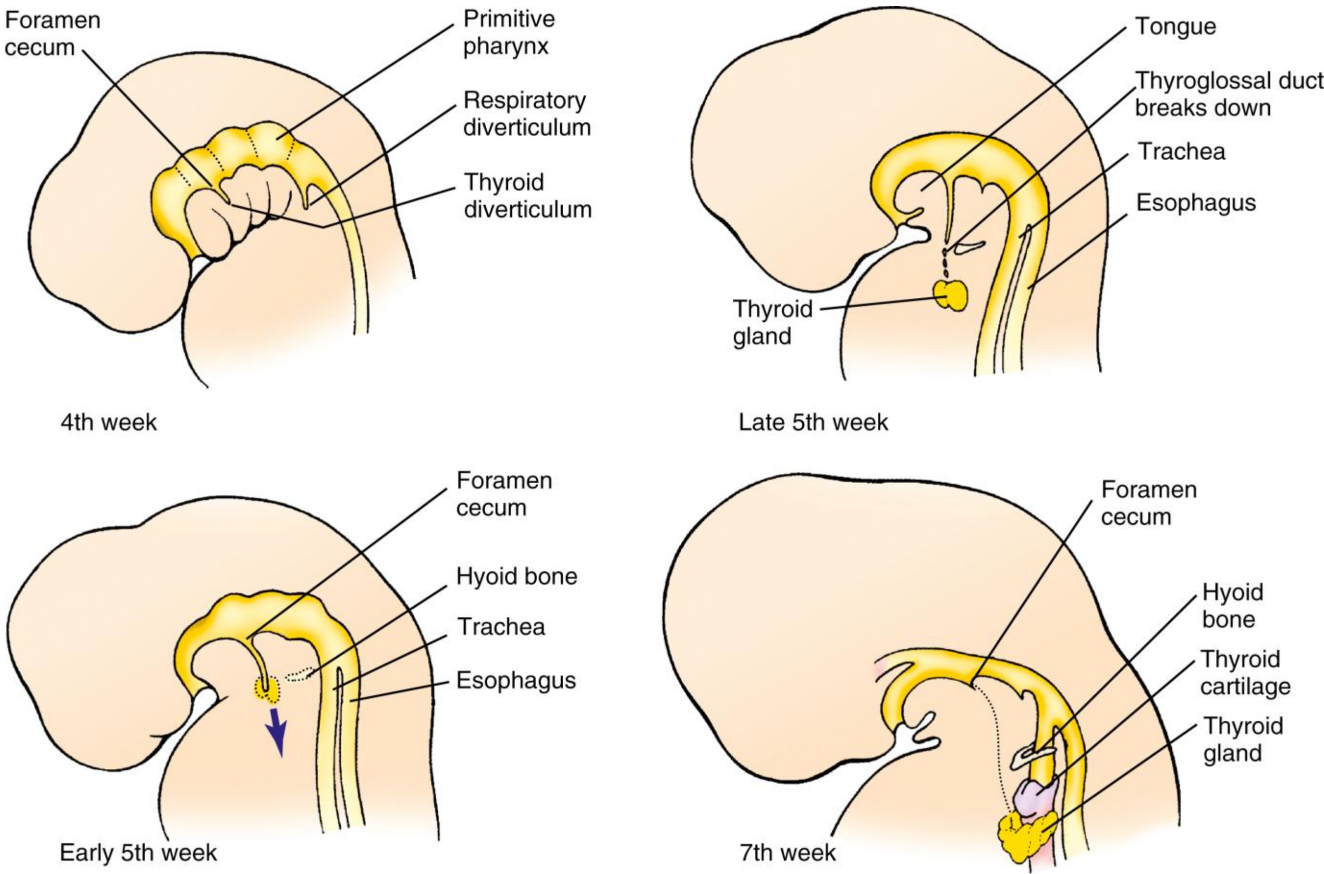
Pierre Robin sequence (1:15 000)

- Mandibular hypoplasia (Micrognathia or retrognathia)
- Glossoptosis (tongue in pharynx - causes airway obstruction and blocks palatal shelves)
- Bilateral cleft palate (Problems with feeding and respiratory distress)

Clinical correlates	Embryological defects
Treacher Collins syndrome	Failure of migration of the neural crest cells into the first arch. Micrognathia, cleft palate, underdeveloped zygoma, conductive hearing loss, malformed pinna, drooping lateral eyelid.
Pierre Robin sequence	Failure of migration of the neural crest cells into the first arch. Micrognathia, glossoptosis, cleft palate
Cervical cyst	Persistence of parts of cervical sinus and /or 2nd pharyngeal groove with no internal or external communication
Cervical fistula	Persistence of the 2nd pharyngeal pouch and cervical sinus. Opening on the side of neck externally & in the tonsillar sinus internally.
Cervical sinus	
External cervical sinus	Failure of the second pharyngeal groove and cervical sinus to obliterate. Open externally along anterior border of SCM
Internal cervical sinus	Failure of the cervical sinus to obliterate and persistence of the proximal part of the second pouch. Opens internally into the tonsillar fossa or near the palatopharyngeal arch

Thyroid Gland Development

- **Endodermal** thickening in floor of pharynx
- Forms an outpouching – **thyroid primordium** – eventually becomes a solid mass
- Descends in the neck anterior to hyoid
- Connected to the tongue at the **foramen cecum** by narrow tube – **thyroglossal duct**
- Thyroglossal duct degenerates and disappear

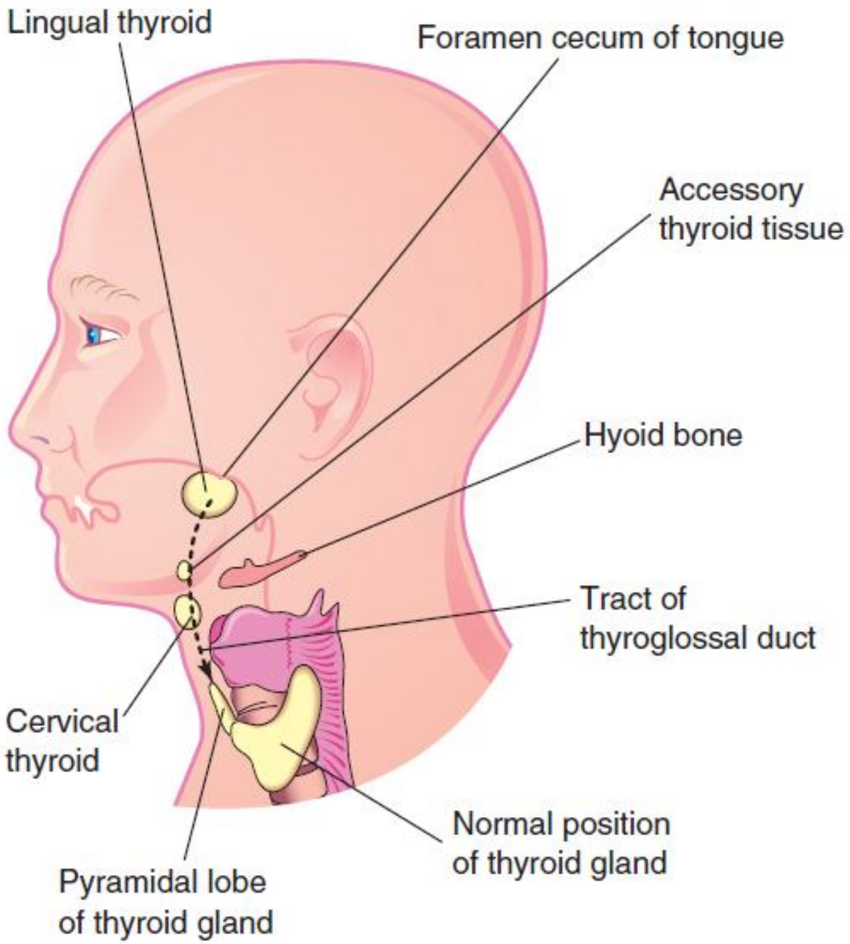


Thyroid Gland Anomalies

- **Ectopic thyroid**
  - **Abnormal position** of thyroid gland
  - Uncommon
  - Occurs along path of descending thyroid gland
  - Lingual thyroid most common

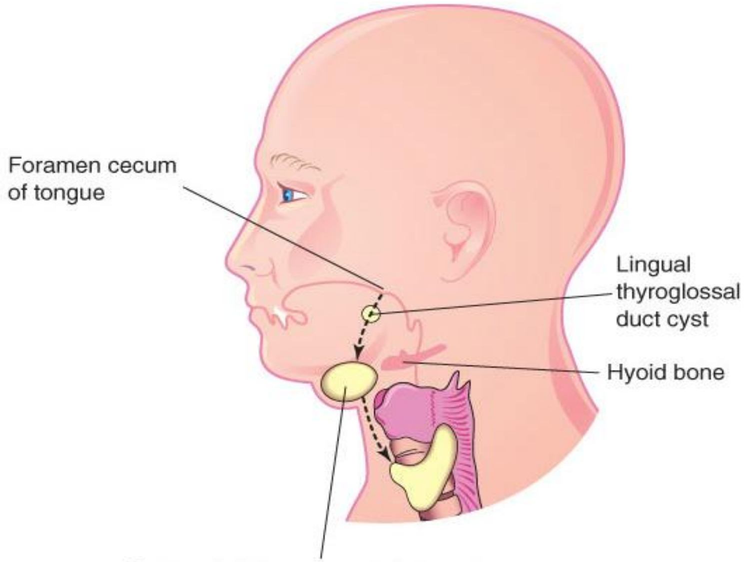
- **Sublingual thyroid** :
  - Incomplete descent
  - High in neck **at or just below hyoid bone**
  - May be only thyroid tissue present

- **Pyramidal lobe** – superior from the **isthmus**
  - Persistent portion of the inferior part of the thyroglossal duct that has formed thyroid tissue

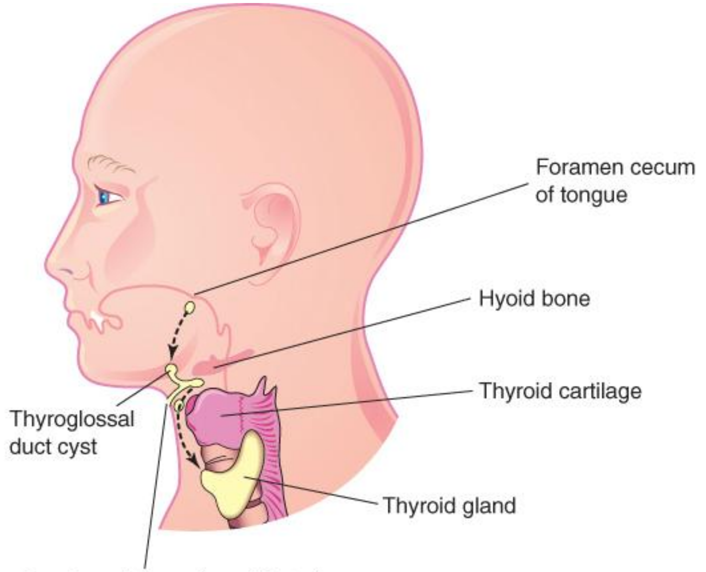


Thyroglossal Duct Cyst and Sinus

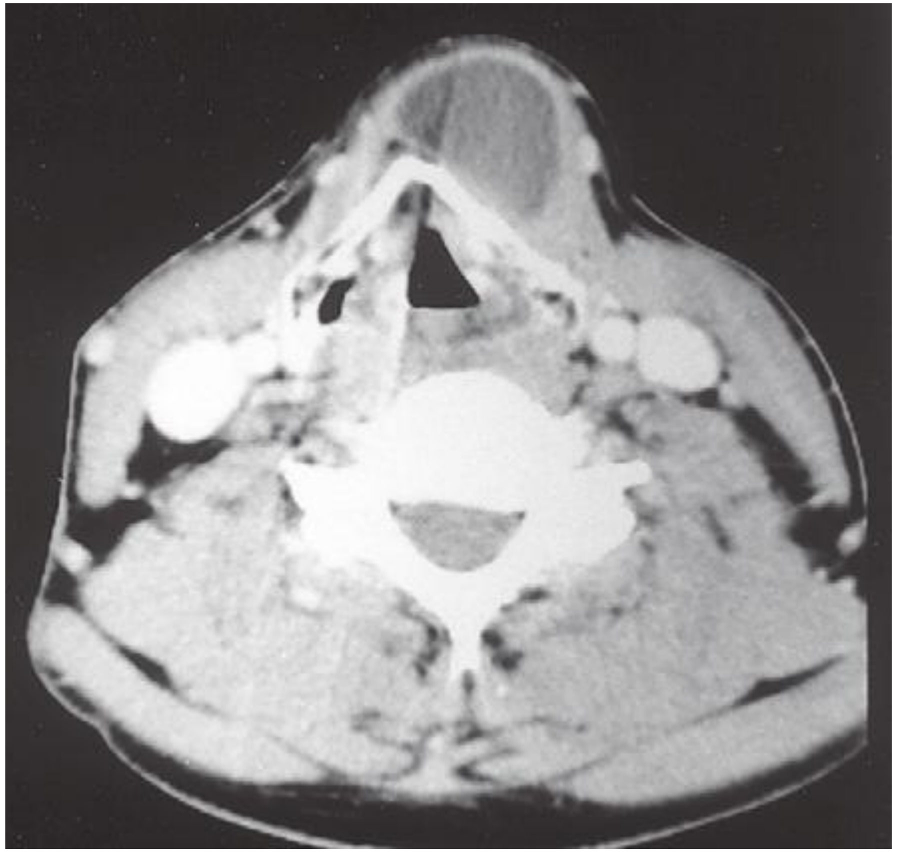
- Cystic remnant of the thyroglossal duct
- Usually in the midline, 50% close to the hyoid bone
- May develop anywhere along the pathway of thyroglossal duct
- Clinically - painless, progressively enlarging, movable **midline** mass
- Infection can cause perforation unto to skin of the neck leading to development of a thyroglossal duct sinus



B Cervical thyroglossal duct cyst



A Opening of thyroglossal duct sinus



Thyroglossal cyst

Identify

- ➢ Bony landmarks
- ➢ Muscle
- ➢ Vessels
- ➢ Pathologies
- ➢ Embryonic origin
- ➢ <u>Anatomical position:</u>
  - Midline
  - Lateral

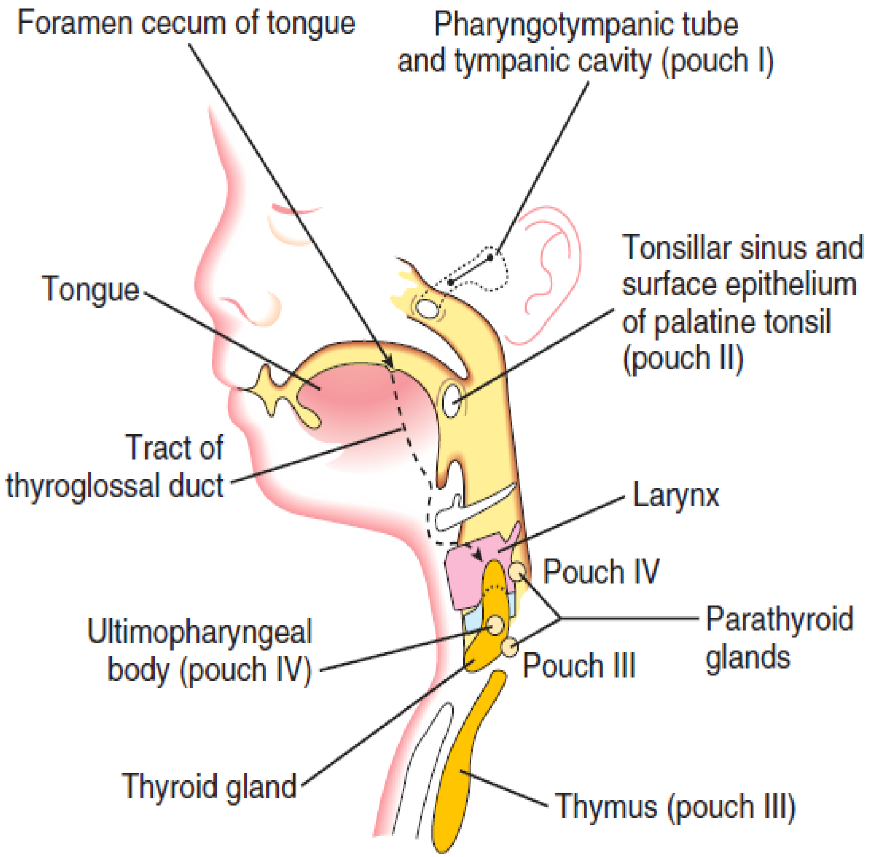
Branchial cyst

Parathyroid Gland

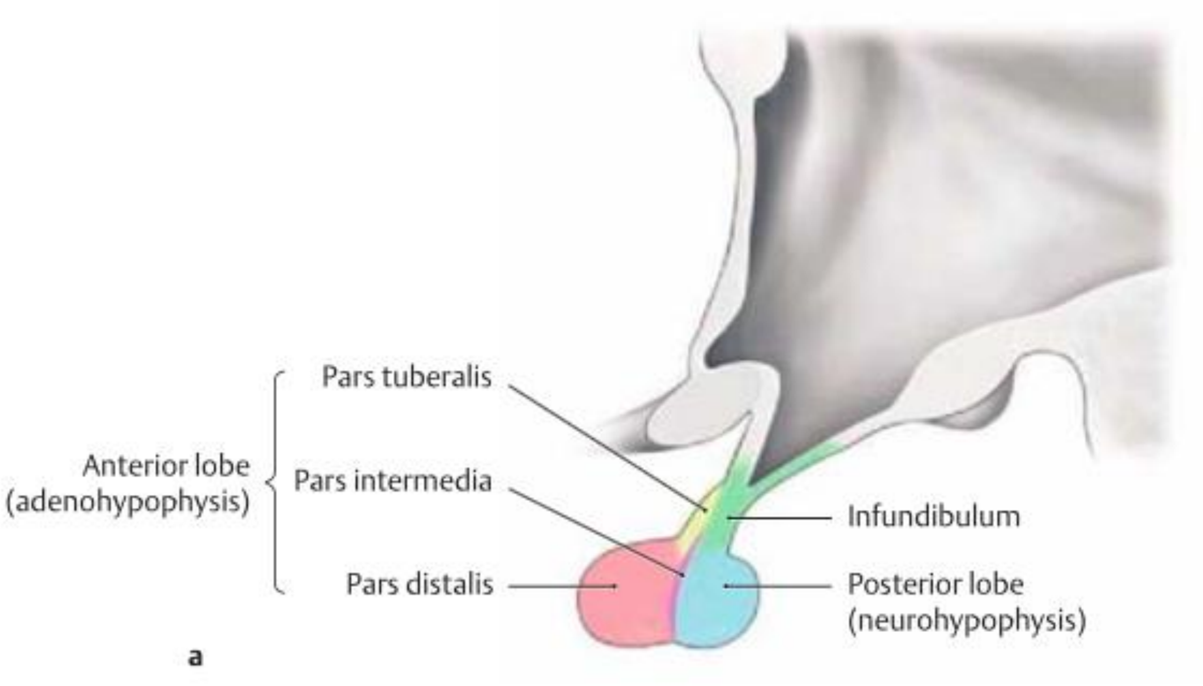
- Develops from proliferation of the epithelium in the dorsal part of the **3rd and 4th pharyngeal pouch** – primordium of parathyroid gland

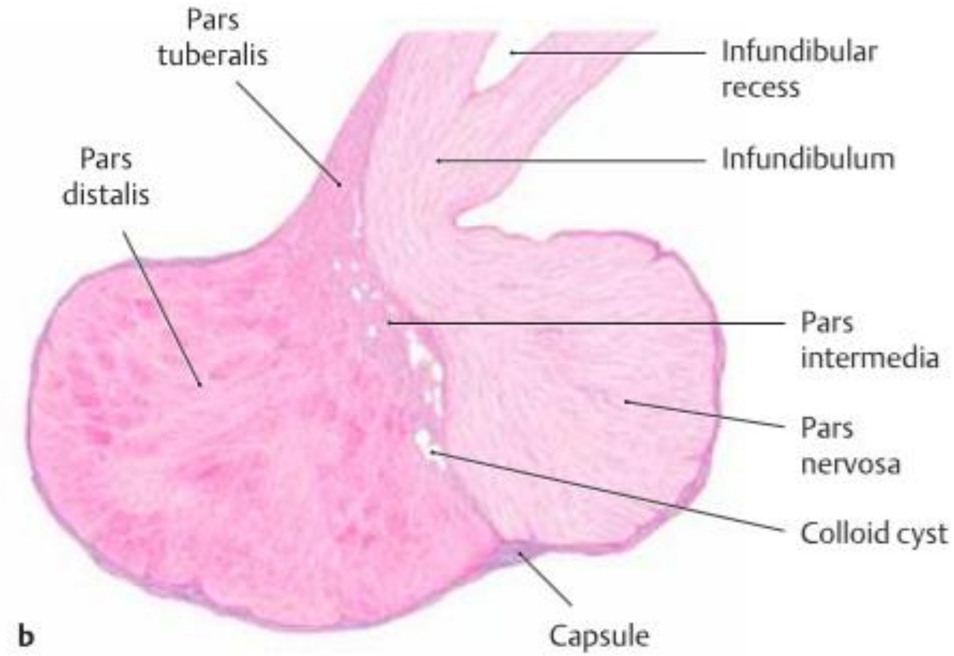
- **Chief cells** function during embryonic life, (oxyphil cells differentiate 5 and 7 years)

- **Anomalies:**
- **Ectopic parathyroid glands** – usually within the thyroid gland or thymus
- **Supernumerary parathyroid glands** – early division of the primordium of the gland
- **Absence of the parathyroid gland** – atrophy or failure of differentiation of the primordium



Pituitary Gland Development





- **Hypophyseal diverticulum,** develops from the **oral cavity,** forms the adenohypophysis (anterior pituitary – glandular tissue)
- **Neurohypophyseal diverticulum,** develops from the developing **forebrain,** forms the neurohypophysis (posterior pituitary – nervous tissue)

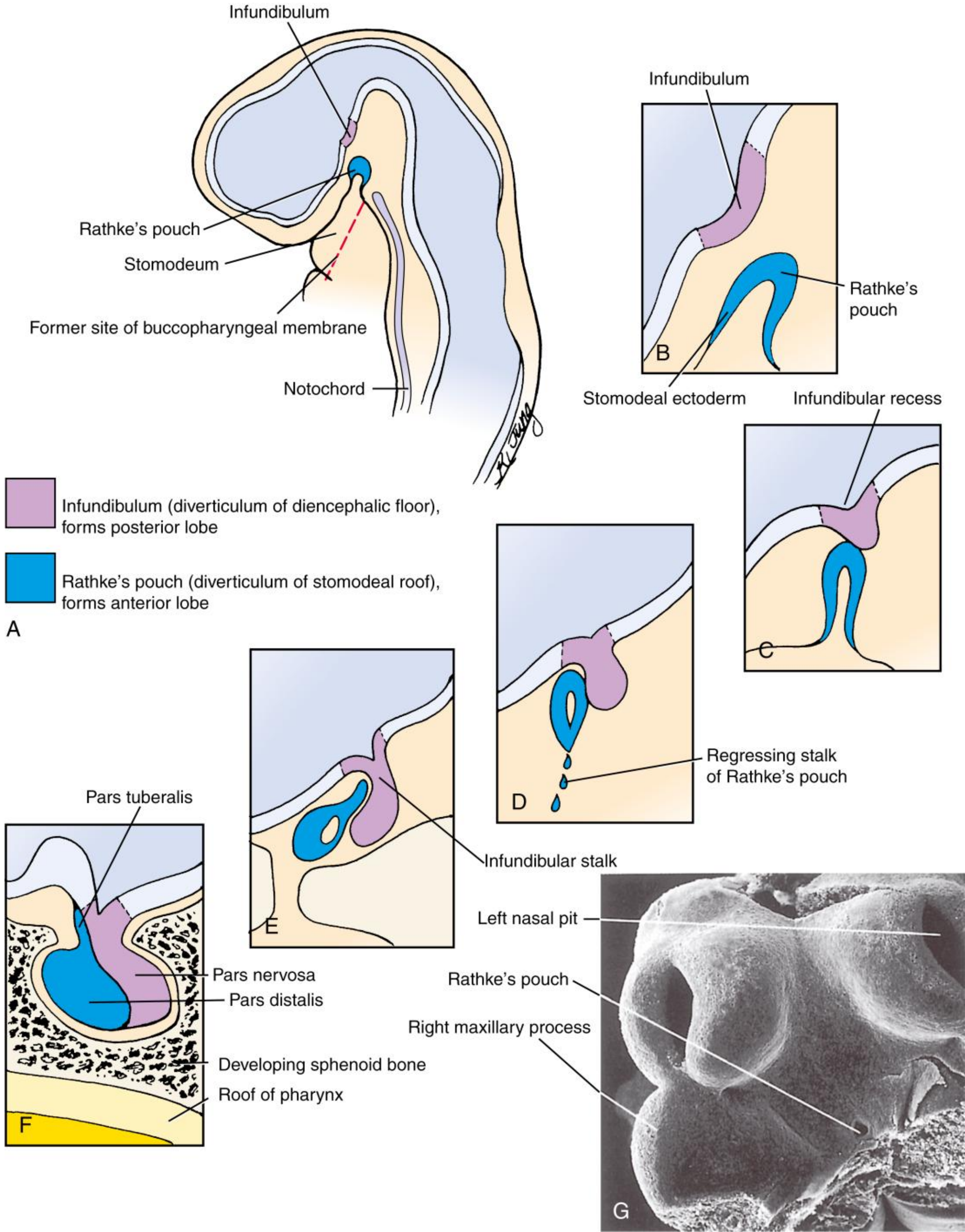
Pituitary Gland Development

- **Hypophyseal diverticulum** (Rathke's pouch) loses connection with the oral cavity to lie in close contact with the neurohypophyseal diverticulum (infundibulum)

- Cells of the anterior wall of Hypophyseal diverticulum pouch increase and form the anterior lobe = **adenohypophysis**

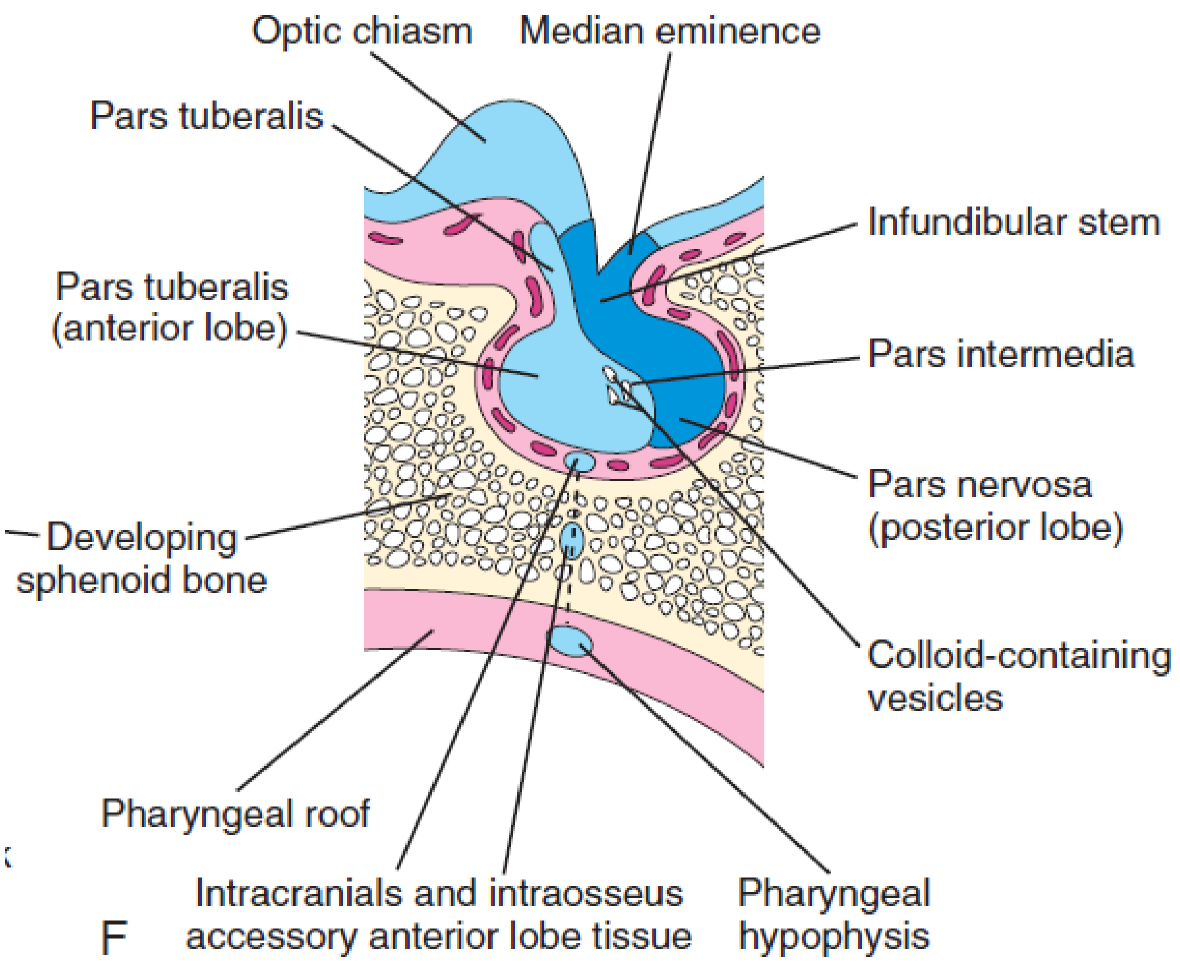
- Neurohypophyseal diverticulum gives rise to the stalk, and the posterior lobe = **pars nervosa**

- A small extension of the anterior lobe grows along the stalk– **Pars tuberalis**



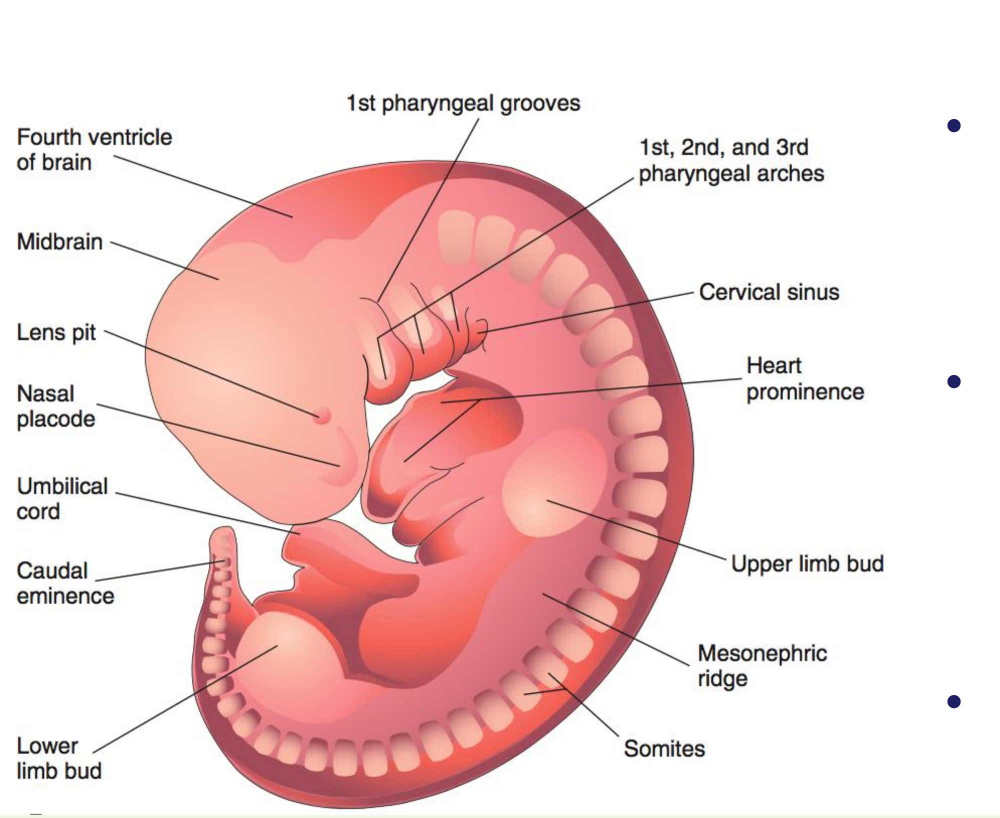
Abnormal Pituitary Gland development

- **Pharyngeal hypophysis: remnant of the stalk** of the hypophyseal diverticulum in the **roof of pharynx**
- **Accessory Pituitary Tissue:** may be **intracranial or interosseous**
- **Craniopharyngiomas:** usually benign and sometimes functional tumors may develop from **minor epithelial remnants** of the hypophyseal diverticulum



Clinical correlates	Embryological defects
Ectopic thyroid	Incomplete descent, abnormal position of the thyroid
Sublingual thyroid	Incomplete descent, high in neck at or just below hyoid bone
Pyramidal thyroid lobe	Persistent portion of the inferior part of the thyroglossal duct that has formed thyroid tissue
Thyroglossal duct cyst	Cystic remnant of the thyroglossal duct
Supernumerary parathyroid gland	Early division of the primordium of the parathyroid gland
Craniopharyngioma	Epithelial remnants of the hypophyseal diverticulum

Cranio-facial development - Summary



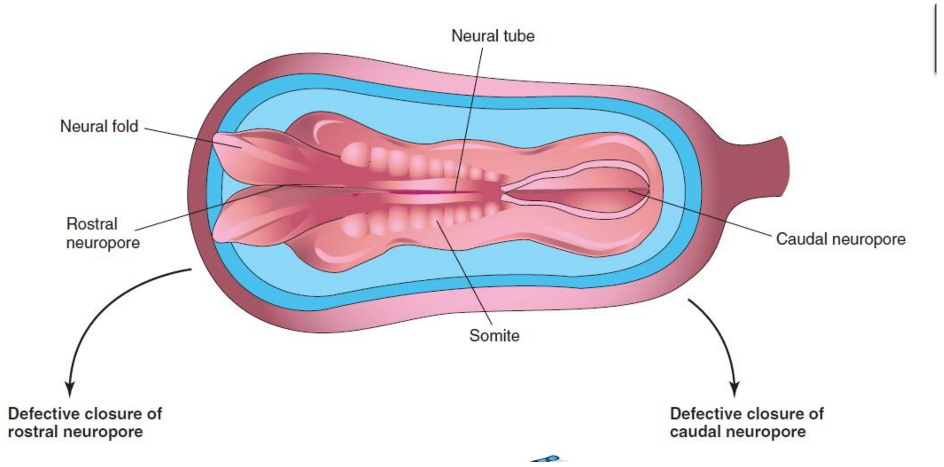
- Skull develops from **mesenchyme** surrounding the developing brain
- **Neural crest cells** migrate from the neural tube to the craniofacial region to form the mesenchyme of the **facial prominences**
- Because the facial prominences are influenced by brain development**, alterations in brain morphology will affect the developing face** and overlying structures
- Neural crest cells are generally responsible for the induction of the skeletal elements of the face

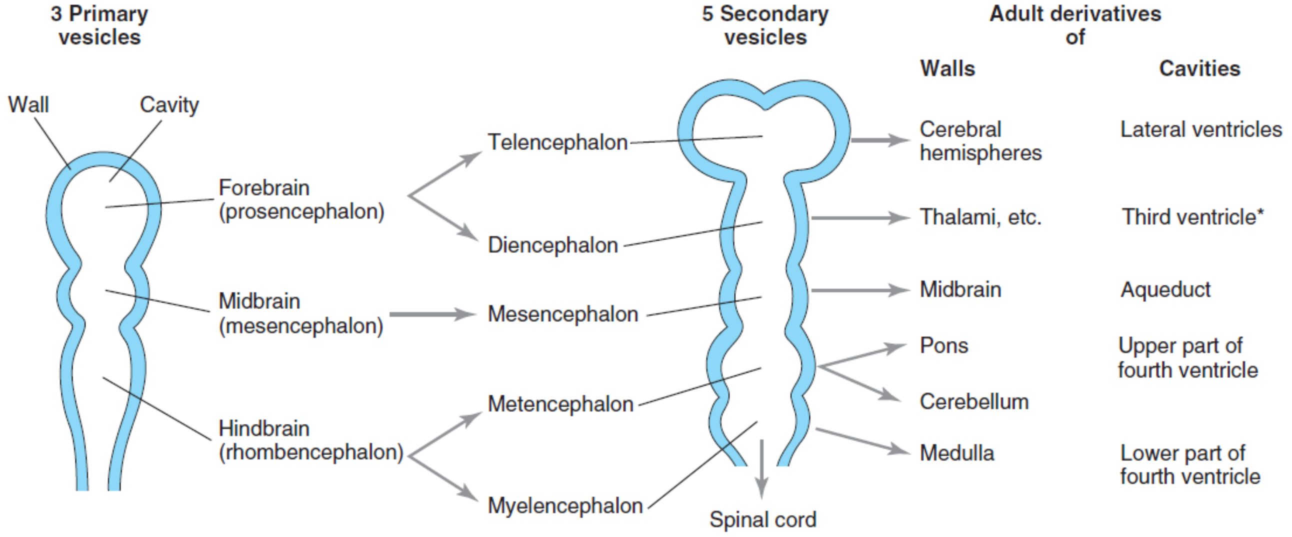
Neural Tube Defects [NTD]

- Neural tube gives rise to brain & spinal cord

- Neural tube defects – critical period **3 – 16 weeks** [by 16th week most of the neuronal development is completed]

- Most common – cranial / caudal neuropores fail to close [Common anomaly in still-born fetuses]





Meroencephaly/Anencephaly

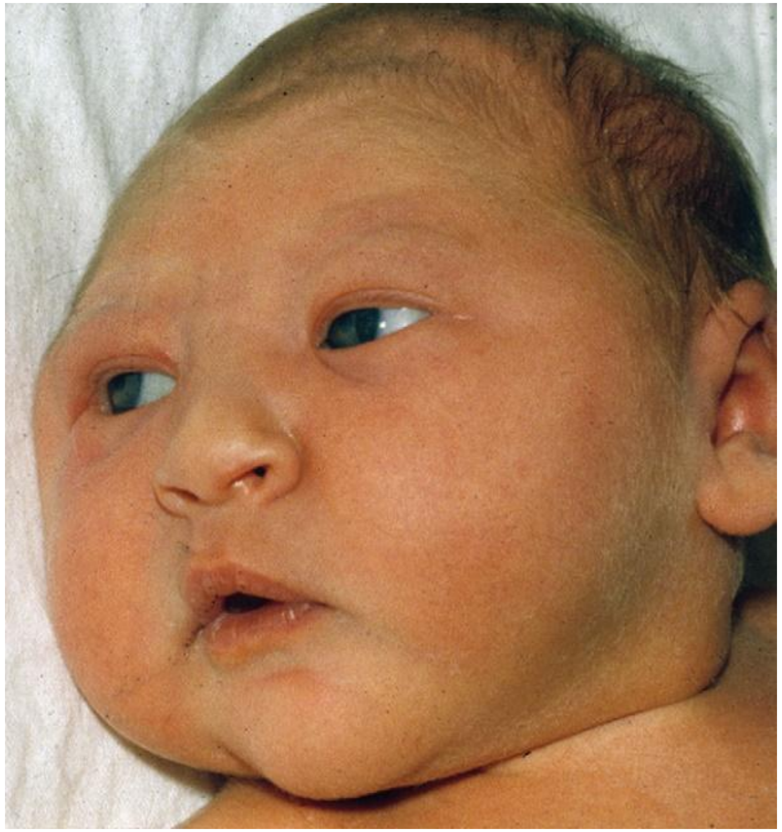
- 1 in 1000 births, common in females
- Cranial neuropore fails to fuse in week 4
- **Skull is not formed**
- Brain tissue (if present) is disorganized and exposed to the amniotic fluid which causes necrosis / degeneration
- Rudimentary brain stem is usually present
- Elevated **alpha fetoprotein**; polyhydramnios

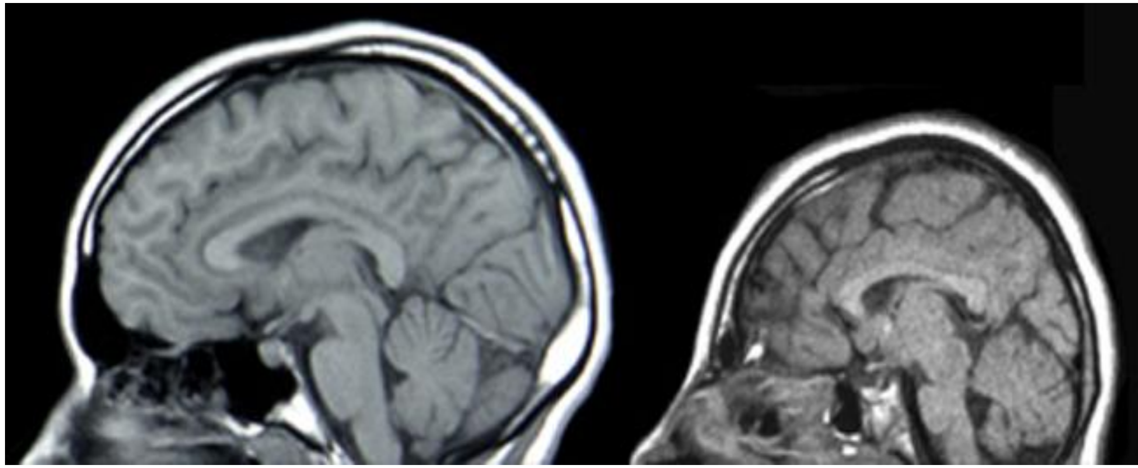
Holoprosencephaly

- 1:16000 live births
- No cleavage of the **prosencephalon**
- Small undivided forebrain and a large single fused ventricle
- Can cause craniofacial anomalies ranging from mild to severe. Usually, **midline structural abnormalities** due to a smaller frontonasal prominence i.e., there might be a single eye, nasal cavity and single incisor teeth
- Due to environmental and genetic factors (**Sonic Hedgehog**)
- Severe NTD Most infants usually die within 6 months

Microcephaly

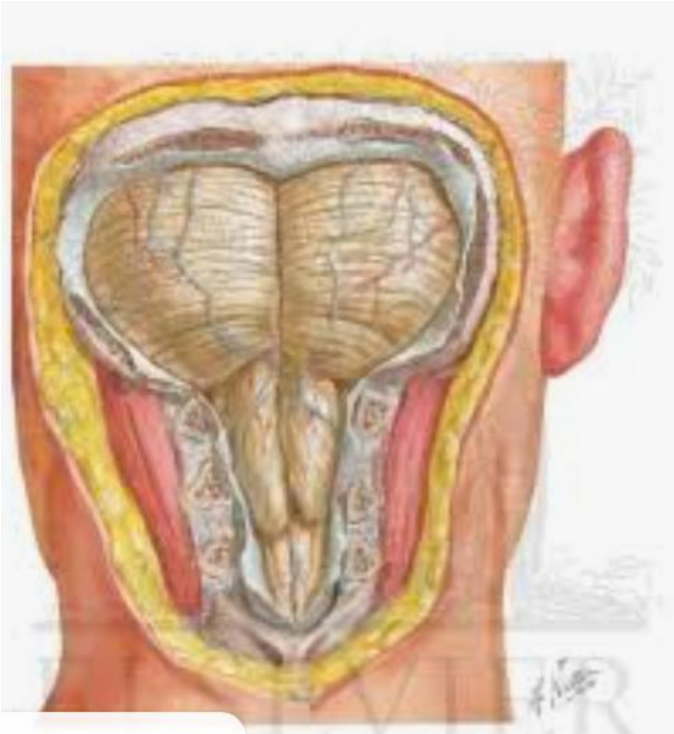
- **Abnormally small brain and calvaria** but normal sized face
- Reduction in brain growth
- Underdeveloped brain leads to severe **neurological defects**
- Genetic and environmental factors
  - Viruses
  - Ionizing radiation
  - Drugs: Fetal Alcohol Syndrome

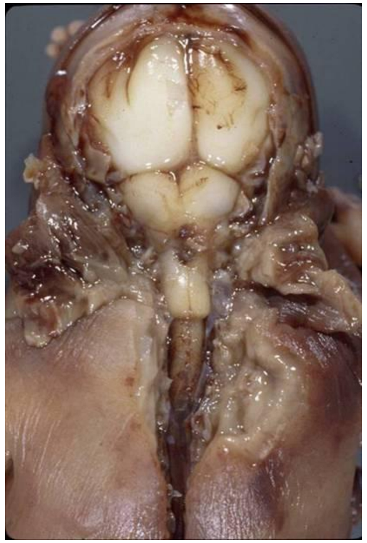


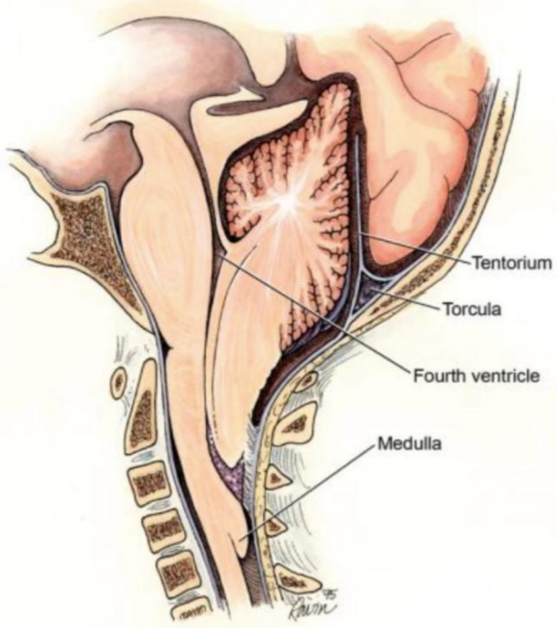


Arnold-Chiari Malformation

- Most common anomaly of the cerebellum (1:1000)
- Inferior displacement of the vermis of the cerebellum and medulla through the foramen magnum into the vertebral canal
- Posterior cranial fossa is small
- Associated with **non-communicating hydrocephalus**
- Cranial nerves **IX, X and XII** may be stretched causing difficulty in speech, swallowing and absent or diminished gag reflex







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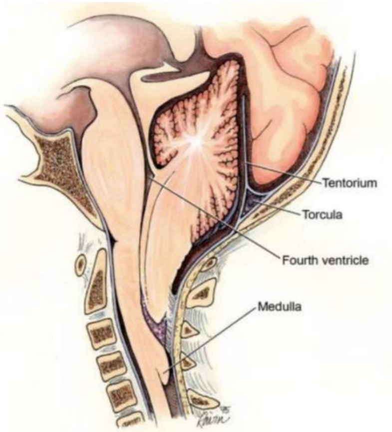
Arnold-Chiari Malformation

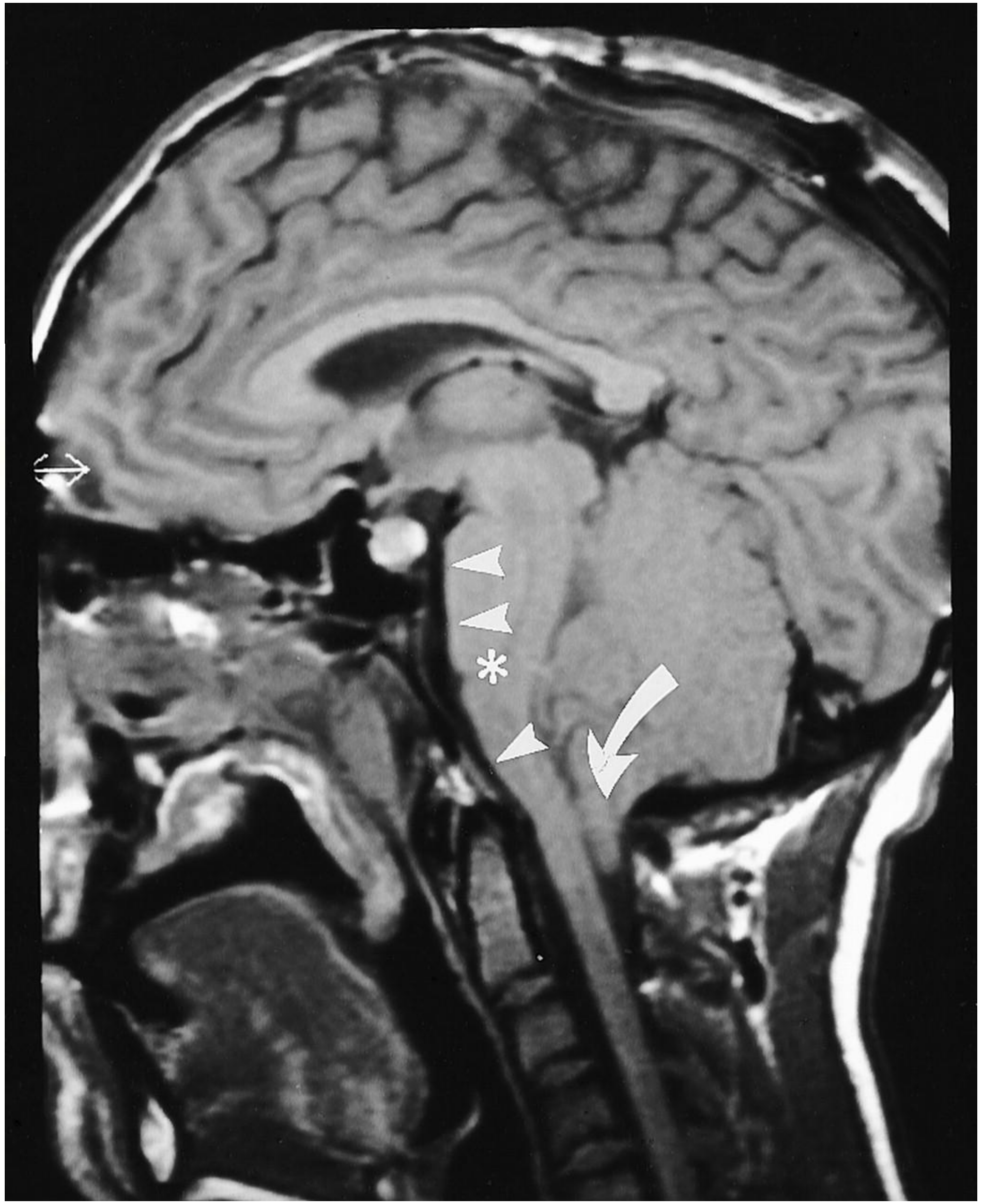
Clinical symptoms:

- Headaches, nausea and vomiting due to raised intracranial pressure
- Muscle weakness in the head and face
- Difficulty swallowing
- Impaired coordination
- In severe cases paralysis



Imaging:

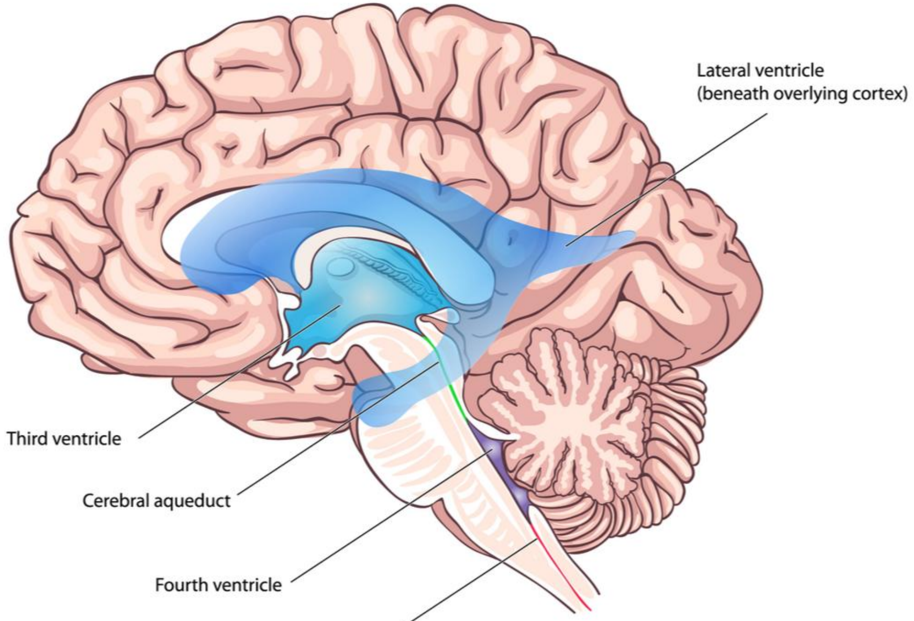
- Downward displacement of cerebellar **tonsils through foramen magnum**
- Compression of the brain stem
- Possible dilation of the ventricular system due to **non-communicating hydrocephalus**

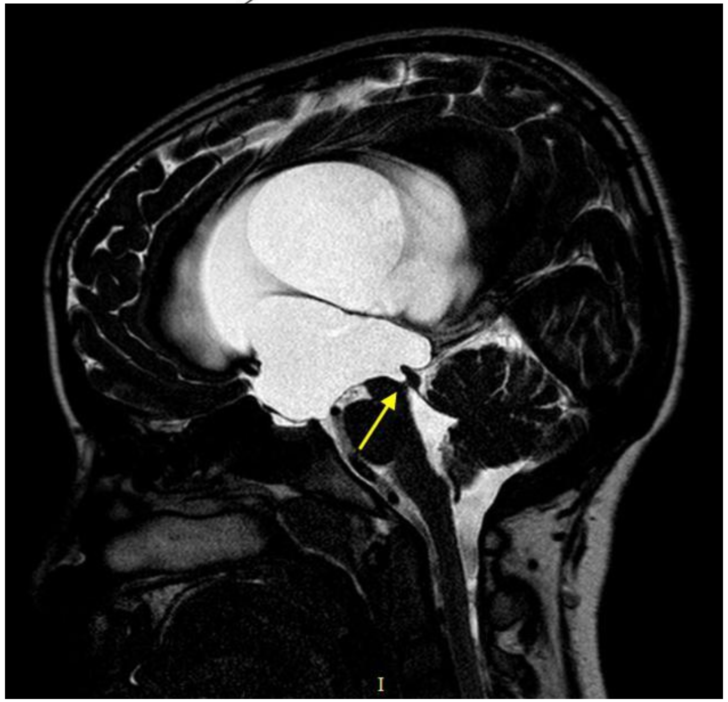


Clinical correlates	Embryological defects
Meroencephaly	Cranial neuropore fails to fuse in week 4
Holoprosencephaly	No cleavage of the prosencephalon
Microencephaly	Abnormally small brain and calvaria but normal sized face
Arnold-Chiari malformation	Inferior displacement of the vermis of the cerebellum and medulla through the foramen magnum into the vertebral canal

Hydrocephalus

- **Non-communicating**
  - **Block** in CSF flow in the ventricular system
  - Congenital stenosis of apertures between the ventricles (aqueduct stenosis is most common cause)
  - Dilatation of ventricles proximal to obstruction
  - Pressure from the accumulated CSF enlarges the head, thinning of the bones and cerebral cortex

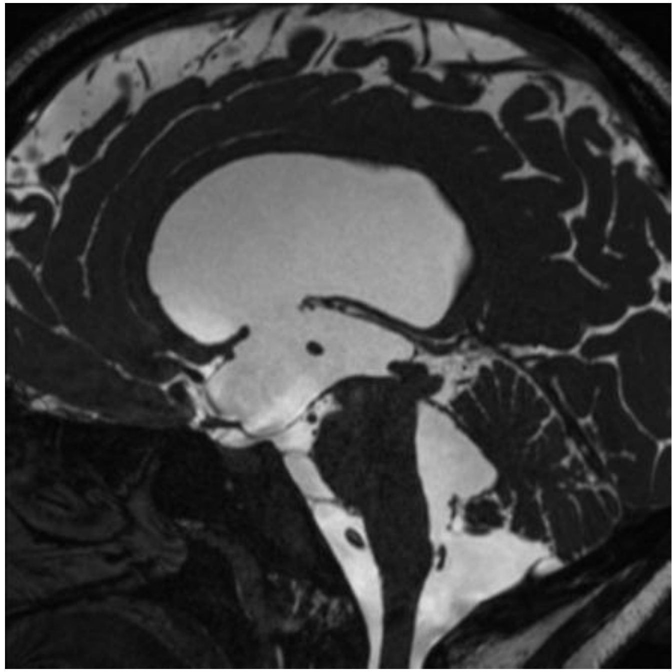




https://radiopaedia.org/articles/obstructive-hydrocephalus?lang=us

Hydrocephalus

- **Communicating**
  - **Reduced absorption of CSF** (due to obstruction of CSF flow through usually at the level of the arachnoid granulations) or **overproduction of CSF** (very uncommon, and only really seen in choroid plexus papillomas)
  - CSF flows in all ventricles, which remain open
  - Common causes include subarachnoid hemorrhage and meningitis
  - Dilation of the entire ventricular system



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